GLOBALTAP: GLOBALLY DIVERSE BEAT TRACKING BENCHMARK VIA CROWDSOURCED TAPPING

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ABSTRACT

Existing beat-tracking benchmarks draw from a narrow range of musical cultures, predominantly Western, and are difficult to scale due to reliance on expert annotation. We address both limits with GlobalTap, a benchmark of 200 popular-music excerpts spanning 59 countries, annotated by 746 participants recruited across ten countries. The benchmark is created using a scalable online tapping pipeline that records taps via audio for high temporal precision, aggregates pooled tap times into consensus beat annotations, and computes intrinsic quality measures that flag perceptually difficult excerpts without requiring a reference. In a separate study on a different 240-excerpt validation set, tapped by an additional 733 participants, an independent group of 130 listeners rated how well the resulting beat tracks lined up with the music, and we detected no overall difference between our annotations and existing reference annotations. Crowd-derived annotations occasionally diverged from the references in phase and metrical interpretation, yet listeners did not consistently prefer the reference versions, suggesting that crowd-derived and existing reference annotations can reflect different perceptual or structural interpretations. The disagreements were often systematic rather than random, and metrical-level choices differed across the ten recruitment countries. We release GlobalTap together with open-source pipeline code as a listener-derived resource for evaluating beat-tracking systems on culturally diverse music.

1. INTRODUCTION

Beat tracking, the automatic estimation of beat times in a musical recording, is a core MIR task, as beat estimates support analysis of tempo, meter, and structure [1]. Recent systems achieve strong accuracy by training on multiple annotated corpora, spanning solo instrument recordings, music with time-signature changes, and classical music with substantial tempo variation [2].

The community currently relies on a small set of established annotated datasets, many of which are consolidated in the Beat This! unified training set [2]. This resource comprises 4,556 annotated tracks from 15 corpora, including Ballroom [3], RWC [4], Beatles [5], and Harmonix [6]. An important limitation of these datasets, however, is that they are heavily biased toward Western popular and dance genres [7]. Expanding genre coverage is difficult because traditional annotation workflows do not scale easily, since they require recruiting musicians with expertise in the relevant musical styles and using specialist software for time-consuming manual annotation [2, 8, 9].

We address this limitation with a scalable online tapping pipeline (Fig. 1A). Participants recruited through standard crowdsourcing platforms tap along to musical excerpts using REPP, an audio-based method for high-precision online sensorimotor-synchronization experiments [10]. Pooled tap times are smoothed into a density over perceived beat locations, and a simple optimization step converts this density into a regularized consensus beat annotation. The approach builds on tapping studies of beat perception [11] and on listener-derived annotations used in MIREX-style evaluation [12], extended to larger and more diverse participant pools [13, 14]. Because multiple participants respond to the same stimulus, the pooled taps also capture variation in perceived beat location and metrical interpretation [15], yielding intrinsic quality measures that do not require reference beats.

We validate the pipeline in a listening experiment over 240 excerpts, 200 of which are drawn from corpora with existing reference annotations (Ballroom, MIREX, and a set of mixed annotated corpora). We use the pipeline to create and release **GlobalTap**, a listener-derived beat-tracking benchmark of 200 popular-music excerpts sampled from 59 countries [16] and annotated by participants recruited across ten countries (Fig. 1B). The release in-

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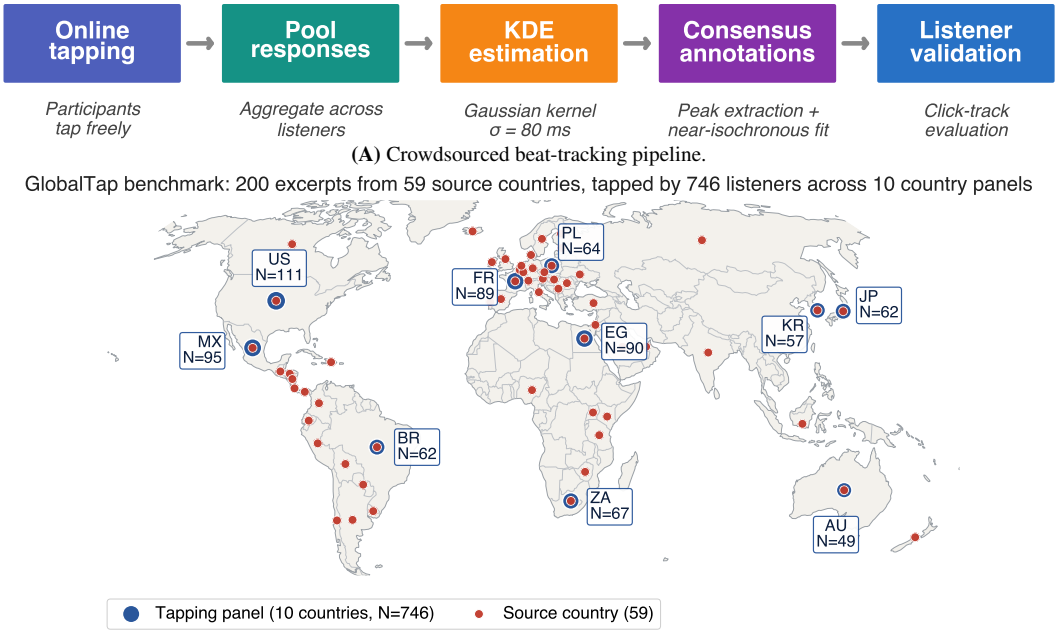

(A) Crowdsourced beat-tracking pipeline.

(B) Geographic distribution of tapping panels and source countries.

Figure 1. GlobalTap pipeline overview and benchmark geography. **(A)** Online tapping responses are pooled across listeners, smoothed by Gaussian kernel-density estimation, and converted into a near-isochronous consensus beat annotation. A separate listening experiment validates the pipeline on click-track overlays. **(B)** The 200 benchmark excerpts are drawn from 59 source countries (red). Tappers were recruited across 10 country panels (blue, $N = 746$ tappers in total). The 10 countries were: United States (US), South Korea (KR), France (FR), Mexico (MX), Egypt (EG), Australia (AU), Brazil (BR), Japan (JP), Poland (PL), South Africa (ZA).

cludes consensus beat annotations, de-identified tap onsets, tap-density summaries, excerpt metadata, intrinsic quality measures, and scripts for reproducing the aggregation and evaluation pipeline. [1]

1.1 Beat Annotation Datasets

Existing beat-annotation corpora differ primarily in how beat times are obtained and how much expert correction is applied. Many established corpora use *semi-automatic expert correction*, in which initial beat estimates from an algorithm, click/metronome track, or bar-level annotation are then checked and corrected by trained annotators. This includes Beatles [5], GTZAN [17, 18], RWC [4, 19], and Ballroom [3]. TapCorrect [20] partly automates this step by snapping tapped beats to nearby audio cues.

Other resources rely on manual expert annotation. Candombe [8] provides expert beat and downbeat annotations, and Harmonix [6] was annotated by professional musicians in a digital audio workstation (DAW).

A related strand derives beat times from tapping or clapping responses. Hainsworth [21] used recordings of a trained musician clapping in time with the music, while Carnatic and Turkish makam resources used expert tapping followed by manual correction [9, 22, 23]. The MIREX beat-evaluation set [12] goes further, recruiting 40 listeners (approximately balanced between musicians and non-musicians) to tap along to musical excerpts, treating each listener's taps as an individual annotation.

Finally, some datasets derive annotations from symbolic or recording metadata rather than from audio. ASAP [24] aligns scores to performances to obtain beat and downbeat times, GuitarSet [25] derives beats from click-track and meter metadata, and Groove MIDI [26] provides tempo-aligned MIDI drum performances.

A key limitation is that most established corpora focus on Western popular, dance, and classical genres, or on solo-instrument and drum-set performance. Non-Western traditions appear only in a small number of expert-built datasets, such as the Candombe [8], Carnatic and Turkish makam [9], and Cretan traditional music [23] datasets used in our validation experiment. Annotator diversity is also limited, with many corpora relying on one or a few expert or musically trained annotators. Such annotations can validly represent structurally informed or culturally situated interpretations, but they do not necessarily show where ordinary listeners hear the beat.

1.2 Tapping as a Measure of Beat Perception

Tapping along with music has long been used in music-cognition research to study beat perception, revealing how tapping behavior is shaped by tempo, meter, and musical structure [11, 27]. Listeners do not simply tap where the onsets are, as perceived beat timing can shift depending on a sound's attack and duration, producing P-center effects [28–30]. Moreover, the same excerpt can support multiple metrical interpretations, so collecting taps from many listeners provides an operational measure of perceptual consensus, while disagreement reveals metrical ambiguity [15]. How easily a pulse is found depends on the music itself, with syncopation and the amount of musical

[1] Code, data, and supplementary HTML: https://github.com/williambotticelli-wells/GlobalTap.

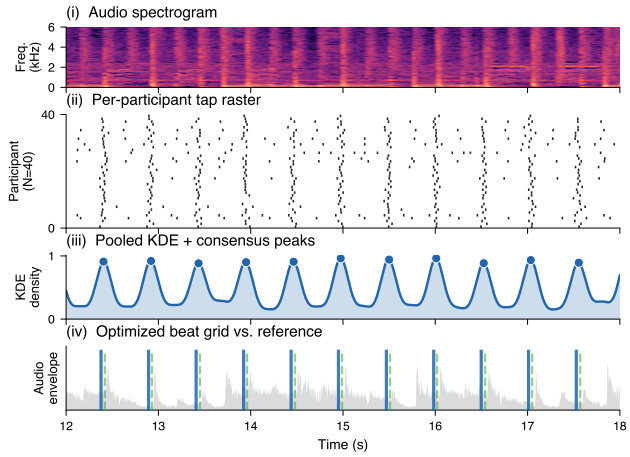

Figure 2. From audio to validated beat grid (Ballroom example, 12–18 s window). **(i)** Audio spectrogram. **(ii)** Per-participant tap raster (N=40 tappers). **(iii)** Pooled tap density (Gaussian kernel density estimate, KDE; σ=80 ms) with extracted consensus peaks. **(iv)** Optimized near-isochronous consensus grid (solid blue) overlaid on the audio envelope (gray) and compared with the expert reference annotation (dashed green).

information predicting pulse-finding ability [31]. Multiple tap sequences can also be used to estimate both the underlying beat times and the uncertainty of the taps around them, without ground truth [32]. Tapping therefore provides a psychologically grounded measure of beat perception, capturing when listeners synchronize with music and the learned biases that shape how they organize rhythmic structure [33, 34].

2. METHODS

2.1 Materials

Our pipeline converts crowdsourced tapping responses into consensus beat annotations in three stages: (a) online tapping data collection, (b) kernel-density aggregation of pooled taps into an empirical distribution over perceived beat locations, and (c) a simple grid-search fitting step that distills this density into a regularized beat track (Fig. 2). The aim is to test whether crowdsourced tapping can support beat annotation with a transparent proof-of-concept estimator, rather than a highly optimized audio-informed model.

The GlobalTap benchmark contains 200 popular-music excerpts of approximately 30 s sampled from the 1,180-song GlobalMood corpus [16] (see Fig. 1B). GlobalMood drew tracks from 2017–2023 weekly YouTube charts across 59 countries, excluding any track charting in more than one. The benchmark was designed to align with our participant-recruitment strategy. Five countries (Mexico, the United States, France, South Korea, and Egypt; see §2.2) were oversampled for stimuli and, on average, for listeners. Each of these countries contributed 20 excerpts, yielding 100 excerpts in total. The remaining 100 excerpts were drawn at random from the other 54 countries, one or two per country, which keeps geographic coverage broad while complementing the five focal countries.

A separate 240-excerpt *validation set* was used to evaluate the pipeline against existing beat references and to test generalization across diverse musical styles. This set comprised 80 Ballroom excerpts [3], 20 MIREX excerpts [12], 100 excerpts from 13 mixed annotated corpora (listed in Appendix A.2), and 40 additional GlobalMood excerpts separate from the benchmark. Reference annotations were available for Ballroom, MIREX, and the mixed corpora, yielding 200 excerpts with paired Crowd and Reference conditions. The 40 GlobalMood excerpts, for which no reference annotations were available, served as an absolute-quality check on globally distributed popular music.

2.2 Participants and Data Collection

Tapping participants. The tapping sample comprised $N = 1{,}479$ participants, 746 for the GlobalTap benchmark and 733 for the validation set, who contributed 36,625 trials and 2,185,514 parsed taps. For the GlobalTap benchmark, tappers were recruited across ten country panels (US, KR, FR, MX, EG, AU, BR, JP, PL, ZA; median 66 per panel, range 49–111), with larger samples on average in the five countries that were also focal source countries for the GlobalMood stimuli (MX, US, FR, KR, EG). Across panels, participants reported relatively low musical expertise (median 1 year of formal training; abbreviated Gold-MSI [35] $M = 4.10$ on a 1–7 scale) and broadly comparable demographic profiles ($M_{\text{age}} = 33.1$; 49.5% female). Recruitment and exclusion counts and further demographic details are reported in Appendix A.3.

Tapping collection. All data collection was conducted online via PsyNet [36], with participants recruited through Prolific (and through Cint for 58 of the 296 GlobalMood validation tappers), under an approved ethical protocol with informed consent. Tapping data were collected with REPP (Rhythm ExPeriment Platform), a browser-based toolkit for sensorimotor-synchronization experiments [10], which records the stimulus and the participant's taps in a single audio stream and recovers tap onsets by aligning that recording to the known stimulus. This bypasses browser timer jitter and gives precise tap timing on ordinary consumer laptops, and it has been used at scale in prior online tapping work [13, 14]. Participants completed audio-volume calibration, a marker-recording check, and practice trials. Those who failed these checks did not proceed to the main task. Trials with failed recordings, failed marker analyses, or incomplete responses were excluded. On each trial, participants heard a ∼30 s excerpt (∼60 s for GlobalMood validation, first 30 s used) and tapped along to the perceived beat. Instructions read: "Keep your taps simple, steady and evenly spaced. Adjust if the tempo changes." We did not assign a metrical level, so participants could tap the pulse they heard rather than one we imposed [15].

Validation experiment. A separate listening experiment validated the resulting click tracks. We recruited $N = 130$ approved participants who contributed 6,487 ratings across the 240 validation excerpts (median 15 ratings per excerpt × condition). Participants rated click-track overlays on a 7-point scale. An instruction asked them to rate how well the clicks line up with the beat of

the music, and the rating question read "How well did the clicks match the beat of the music?" We chose this wording because it is simple and does not point participants toward a specific metrical level or a "correct" answer, and is similar to the Beat Alignment Test [37]. Participants judged the rendered beat sequence as a whole and gave one overall rating, so this design cannot determine whether a score reflects metrical level, phase, local timing, or their combination. Two conditions were compared per stimulus: *Crowd* (beat track produced by our pipeline from consensus tapping; see §2.3) and *Reference* (the reference annotation for each corpus). For Ballroom [3] and the mixed corpora these were the corpora's own expert annotations. For MIREX, we produced a consensus sequence by running our pipeline on the 40 MIREX lab annotators' tap sequences, so that any Crowd–Reference difference on MIREX reflects *tap source* (online crowd vs. lab listeners) rather than how the taps were aggregated. Each excerpt's Crowd and Reference versions were presented in randomized order within the same block. A multiple-choice attention check at the end of the task asked participants what had been added on top of the music, and identifying "short click sounds overlaid on the piece" was required to finish.

2.3 Consensus Pipeline and Rendering

Aggregation. Fig. 2 illustrates the aggregation, optimization, and reference-comparison steps on one Ballroom excerpt. We pooled individual tap times across participants and computed an initial kernel density estimate (KDE) on a 5 ms grid using a Gaussian kernel with $\sigma = 80$ ms. Before pooling, we cleaned each participant's trial by removing a tap when either neighboring inter-onset interval (IOI) exceeded the trial's median IOI by more than 3.5 scaled median absolute deviations (MADs). The remaining taps were pooled to compute the KDE. Peaks were extracted with prominence ≥ 0.10 of the KDE maximum and a 150 ms minimum inter-peak distance [38]. The bandwidth value of $\sigma = 80$ ms was selected during pipeline development, and values between 70 and 100 ms gave comparably stable period and beat-count behavior.

Grid optimization. We used an exhaustive grid-search heuristic designed for short excerpts. An independent group of 130 listeners later rated its beat grids on par with the reference annotations (§3.2). It fits one near-isochronous beat grid $\mathbf{b} = \{\phi + nT\}$ to the extracted peaks $\mathbf{p}$ in four steps (settings in Table A6). First, a grid search over period T (48–214 BPM) and phase ϕ maximized $\mathcal{S}(T, \phi) = \bar{D}(\mathbf{b}) + 0.5\,C(\mathbf{b}, \mathbf{p})$, which rewards beats that land on peaks ($\bar{D}$) and grids that leave few peaks unmatched (C). Here $\bar{D}$ is the mean, over the grid beats, of a curve with the same 80 ms Gaussian at each peak, so every peak counts equally however many people tapped it, and C is the proportion of peaks near a grid beat. Second, the best grid was refined by least squares on nearby peaks, and each beat near a peak moved part of the way toward it, so the beats follow the taps while staying close to evenly spaced. Third, the beat rate was doubled when peaks also fell at most of the midpoints between beats, and gaps were filled at the grid spacing. Fourth, when the peaks formed two groups at different phases, a grid was also fit to each group, and the three grids were compared with a second score that also rewards even beat spacing. The grid spans the whole excerpt, including stretches with few taps, while ratings and tracker scores use only the 10–25 s window. These settings were chosen during pipeline development on the validation corpora (Ballroom, the mixed corpora, and MIREX), then held fixed for all reported analyses, including those of the GlobalTap corpus. To check how much the output depends on them, we varied each setting one at a time by 18–50% (e.g. the grid resolution, the density/coverage weighting, the snap tolerance, or the bimodal-detection threshold) across all 240 validation excerpts and found that outputs changed little, matching the original outputs at F-measure .947–1.000, with the tactus (the metrical level the grid locked onto) unchanged in 92.1–100% of cases, and agreement with the reference annotations changed by at most .014.

Sample size. We assessed convergence with three measures: the correlation between participant-normalized KDEs (density correlation), the F-measure between their extracted peaks ($\pm$70 ms, the standard beat-tracking tolerance [39]), and the F-measure between the final beat grids. Across all 200 benchmark excerpts, two independent groups of $N = 50$ tappers achieved a mean density correlation of $r = .814$ and a peak F-measure of .880, and their optimized beat grids agreed at $F = .918$. Robustness analyses in Appendix A.4 evaluated smaller samples by drawing two independent groups of participants per excerpt and running the complete pipeline on each. At $N = 20$, the two groups' beat grids agreed at $F = .893$. Every validation excerpt had more tappers than that, at least 34 (median 40 for Ballroom, 36 for the mixed corpora, 67 for MIREX, and 110 for GlobalMood validation, see Appendix A.3). For the final benchmark we recruited 114–120 tappers per excerpt (median 117), the maximum feasible within the data-collection period.

Rendering. For the listening experiment, beat times in the 10–25 s window of each excerpt were converted to 1100-Hz, 12-ms Hann-windowed clicks and mixed with the original mono 44.1-kHz audio at music 0.35 / click 0.65, with the music loudness-normalized to -23 LUFS and the clicks at a fixed peak level.

2.4 Intrinsic Quality Measures

We report two reference-free indicators of annotation quality. Beat-sequence *CV* is the coefficient of variation (SD/mean) of the inter-onset intervals of the final beat track, a measure of grid *anisochrony* that can be computed for both Crowd and Reference annotations. *ISTC* (inter-subject tapping coherence; [40]) summarizes agreement among pre-optimization participant taps following the bin-counting definition of Singer et al. For each excerpt we count how many *unique participants* tap in each 100 ms bin (advanced in 50 ms steps), take the largest count in each 1 s window, and average those maxima over the $[0, 30]$ s window. Because unique-participant counts are bounded by panel size, ISTC's absolute scale varies across corpora, so we report it within-corpus. Both ISTC and CV are com-

Table 1. Per-corpus mean listener ratings (1–7 scale) for the pipeline (Crowd) vs. Reference annotations on the 200 paired validation excerpts plus the 40-excerpt GlobalMood absolute-quality set. $\Delta = \text{Crowd} - \text{Ref}$; effect sizes are paired-sample Cohen's d_z; p-values from Wilcoxon signed-rank on excerpt-level means.

Corpus	n	Crowd	Ref	Δ	p	d_z
Overall (paired)	200	4.67	4.68	−0.01	.231	−0.02
Ballroom	80	5.01	5.14	−0.13	.007	−0.22
Mixed corpora	100	4.47	4.37	+0.09	.546	+0.10
MIREX	20	4.33	4.41	−0.08	.557	−0.09
GlobalMood	40	4.93	—	—	—	—

puted over the whole excerpt, while ratings and tracker scores use the rendered 10–25 s window. ISTC is defined only for crowd-derived annotations, because the Reference condition consists of a single annotated grid rather than a set of taps.

3. RESULTS

3.1 KDE–Reference Alignment

As a first check, we asked whether the pooled tapping density already aligns with the reference beat structure before any grid postprocessing is applied. Averaged over excerpts, 90.3% of raw KDE peaks fall within ±70 ms of a reference beat on Ballroom, 56.2% on the mixed corpora, 74.2% on MIREX, and 71.7% overall. Agreement before grid fitting was lowest on the mixed-corpus excerpts dominated by rubato (ASAP) or dense syncopation (Candombe). The pooled tapping signal thus already carries much of the temporal structure that the grid fitting then regularizes.

3.2 Overall Listener Preference Validation

To test whether the optimized consensus beat track is perceived as a viable alternative to expert annotations, we compared the pipeline's crowd-derived track (Crowd) against each excerpt's reference annotation. Per-excerpt mean ratings on a 1–7 scale (see Table 1) were compared with Wilcoxon signed-rank tests on excerpt-level means, with $\Delta = \text{Crowd} - \text{Ref}$ as the effect direction.

Table 1 summarizes the results. Across the 200 paired excerpts, we found no overall difference in ratings between the two annotation types ($\Delta = -0.01$, 95% CI $[-0.12, +0.10]$, $p = .231$, $d_z = -0.02$), indicating no systematic listener preference for either annotation. Ratings also did not differ significantly within the Mixed or MIREX corpora. Reference annotations received significantly higher ratings in the Ballroom corpus, but the effect size was small. These results suggest that our Crowd-based annotations, as judged by the independent raters, are comparable in quality to the references.

The 40 reference-free GlobalMood excerpts received plausible ratings ($M = 4.93$), comparable to those observed for the other corpora. Mean ratings across corpora ranged from 4.33 to 5.14 on the 7-point scale.

3.3 Intrinsic Quality Measures Correlate with Listener Ratings

Both reference-free measures were associated with ratings, supporting their use as reliability indicators. The Crowd annotation's beat-sequence CV correlated with both Crowd (Spearman $\rho = -.43$, $p < .0001$) and Reference ratings ($\rho = -.39$, $p < .0001$) across the 200 paired excerpts, so temporally irregular excerpts seem harder to annotate, whatever the method. Within individual corpora, CV also predicted Crowd ratings for the mixed corpora ($\rho = -.53$, $p < .001$) and Ballroom ($\rho = -.35$, $p = .0017$), but not for GlobalMood ($\rho = -.23$, $p = .15$) or MIREX ($\rho = -.02$, $p = .94$). Pre-optimization ISTC correlated positively with Crowd ratings in the three larger corpora: mixed ($\rho = .68$, $p < .0001$), Ballroom ($\rho = .40$, $p = .0002$), and GlobalMood ($\rho = .52$, $p = .0006$), but not for the smaller MIREX set. All significant associations remain significant after Benjamini–Hochberg correction.

3.4 GlobalTap Benchmark Analyses

On GlobalTap, pooled crowd consensus from all 746 benchmark tappers (§2.2) agreed with madmom [41] at $F = .897$ and Beat This! [2] at $F = .841$ (±70 ms), and tracker–tracker agreement was $F = .875$. These values and the metrics below use the 198 excerpts evaluable for both trackers (Beat This! returned fewer than two beats for two excerpts). Crowd–tracker agreement was thus comparable to tracker–tracker agreement, supporting GlobalTap's direct use as a listener-derived evaluation reference. Using the GlobalTap consensus as the reference, we also computed the standard beat-tracking evaluation metrics [42], including correct metric level (CML) and any metric level (AML, which credits double, half, and offbeat interpretations), each over the longest continuously correct stretch (CMLc/AMLc) or all beats (CMLt/AMLt), along with information gain, which measures how far the spread of beat errors departs from uniform. All were computed with mir_eval defaults [39]. Detailed results are reported in Table A5. Both methods achieved high overall scores, although madmom systematically outperformed Beat This! on all five metrics. Most of madmom's advantage disappears when other metrical levels and offbeat interpretations are accepted (CMLt .807 vs. .636; AMLt .956 vs. .935), so it lies mainly in which level and phase each tracker chose.

3.5 Exploratory Country-Panel Analyses

For each excerpt, we compared each trial's tapping period with the median period across the other nine panels (Fig. 3). Metrical-level selection differed significantly across the ten recruitment panels overall ($p < .001$). For example, Japan showed the least double-rate and most half-rate tapping of any panel (7.5%/20.1%), while Egypt, Mexico, and South Africa tapped at double rate most often (28.1%, 28.0%, and 26.0%), a difference to explore in future work. Tap timing (phase) showed no consistent differences across panels ($p = .152$). In the five focal countries, tapping coherence was not reliably higher on their own country's excerpts (+0.07 ISTC units, 95% CI $[-0.03, +0.16]$, $p = .18$). Note that we grouped partici-

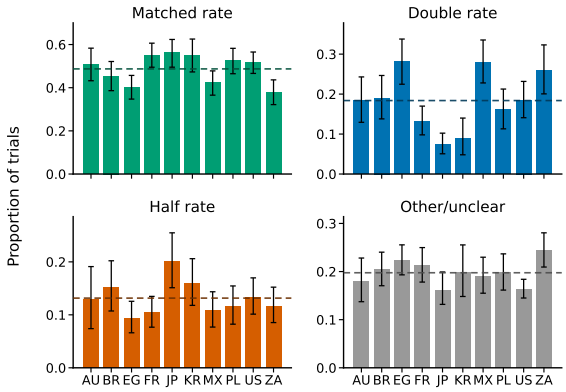

Figure 3. Metrical-level choice by country panel. Proportion of trials whose tapping rate matched, doubled, halved, or otherwise diverged from the median of the other nine panels (95% cluster-bootstrap intervals, dashed line is the mean across panels).

pants by where they were recruited, which may not reflect the music they are enculturated to, and we did not measure how familiar they were with individual excerpts.

3.6 Structured Divergences Between Crowd and Reference Annotations

When the pipeline diverged from expert annotations, the differences were often systematic and musically interpretable. On the 180 excerpts with paired expert beat references (80 Ballroom and 100 mixed-corpora, with the 20 MIREX excerpts excluded because their reference is itself a crowd-pipeline output), three recurring patterns emerged. An example of each is in the supplementary HTML.

Phase offsets. Consistent offsets ≥ 20 ms appeared on 71 of the 145 excerpts where the crowd and reference beat periods matched. On 8 of them the crowd's beats fell between the reference beats, a third to half a beat away. On the other 63, the crowd preceded the reference in 49 cases (up to 144 ms). This is consistent with negative mean asynchrony, P-center effects, or attending to different instrumental or rhythmic layers, though our design cannot separate them [8, 11, 28, 29].

Metrical alternatives. On $35/180$ excerpts, the crowd annotation's period differed from the reference's: 9 at half rate, 14 at double rate, and 12 at other ratios (six near 3:1, 1:3, or 2:3). The nine half-rate cases all had fast reference beats (median IOIs .28–.37 s) while the crowd tapped near preferred rates (.57–.75 s), consistent with listeners favoring a slower level at fast tempi [15]. All nine also received higher Crowd than Reference mean ratings (median $\Delta = +.75$; $+.60$ for the five Ballroom excerpts); double-rate and other cases showed no consistent direction (full table in Appendix A.6). On the 145 excerpts with matching periods, the reference was rated slightly higher (median $\Delta = -.14$, Crowd higher on $51/145$).

Bimodal pooled tapping. On $31/180$ excerpts, the KDE also showed a pulse at half or double the crowd annotation's rate, with at least 40% as many inter-peak intervals at that rate as at the annotation's, indicating that the pooled taps contained two metrical levels before the optimizer selected one grid.

4. DISCUSSION

This study introduces GlobalTap, a fully reproducible benchmark of 200 excerpts from 59 countries, and shows that online crowd tapping provides a scalable approach to beat annotation. Validation analyses support its perceptual validity: overall, listeners rated crowd-derived annotations as comparable to existing references, and beat-sequence CV and ISTC predicted these ratings. Crowd tapping therefore offers a practical complement to expert annotation, which remains valuable but is costly, requires specialist knowledge, and is difficult to scale across diverse musical contexts. Systematic differences between crowd-derived and reference annotations were more consistent with perceptual ambiguity or alternative interpretations than with annotation failure. How listeners perceive and evaluate rhythm also depends partly on familiarity with culturally specific musical styles [14, 43–45], which motivates two features of our approach: it allows us to recruit participants in countries the music was sampled from, and it focuses on non-expert listeners, whose interpretations may differ meaningfully from those of experts, for example in the choice of tactus level [46], and potentially better reflect ordinary listening. Neither source replaces the other, and non-expert listeners may still miss structure that an enculturated musician would identify.

Limitations and future directions. The dataset introduced here remains limited in its geographic coverage and in the scale and density of its annotations. With only 200 excerpts, it is currently better suited for benchmarking than training deep-learning models. However, the pipeline is scalable, and the dataset could be expanded by recruiting further through platforms such as Prolific and Cint to reach a wider set of countries [47].

A further limitation is that our current beat-extraction procedure relies on simple heuristics. For example, it does not accommodate music with substantial tempo fluctuations, such as rubato, or more complex forms of rhythmic organization. This limitation could be addressed in future work, because tapping data can reflect tempo changes provided that those changes are sufficiently predictable [48]. The extraction procedure could also be extended by incorporating acoustic information. Although we did not use acoustic features here, annotations derived from tapping can be refined by aligning them to audio cues [20], and audio-based trackers [2,41] could supply those cues, sharpening timing while leaving the metrical level and phase to listeners.

Despite these limitations, the approach offers a promising means of addressing important gaps in the literature. It could be used to investigate the psychological and cultural factors that shape listeners' choices of tactus and meter, while intrinsic quality measures could provide complementary insight into the difficulty or complexity of metrical structure. Because recruitment platforms can screen for musical training, the same pipeline can also collect annotations from musicians, non-musicians, or mixed groups and compare them directly. The paradigm could also be extended to produce high-quality annotations for challeng-

ing musical materials that remain beyond the capabilities of current state-of-the-art methods, for example by selectively recruiting larger numbers of tappers for tracks where tappers agree poorly with one another or with current trackers. The same framework could also support the collection of other timestamped MIR annotations, including downbeats and section boundaries. Taken together, we view this work as a step toward the broader inclusion of popular musical materials from around the world in beat-tracking research and MIR more generally.

Ethics Statement

We conducted this study in accordance with ethical best practices. All participants, recruited via Prolific or Cint, provided informed consent under Cornell University IRB protocol IRB0150361 ("The cultural foundation of perception and cognition – online experiments"), were compensated fairly according to the platforms' guidelines, and had their data protected by standard safeguards. No personally identifying information is released. The participant data released are limited to de-identified tap onsets, tap-density summaries, derived consensus annotations, per-excerpt mean ratings, and intrinsic-quality measures. Full Global-Tap benchmark source recordings are not redistributed and must be obtained from their source corpora and YouTube, as documented in the repository (`data/SOURCES.md`). Cross-cultural research requires careful attention to a wide range of ethical considerations [49,50], and these concerns informed the design of the present study. We acknowledge that training and evaluation datasets may contain biases introduced through participant recruitment, sampling, and data-selection procedures. Nevertheless, our study aims to address important limitations and societal risks in current MIR technologies, which remain disproportionately focused on Western music and Western participants [7]. Despite these efforts, our benchmark panels were recruited through a single crowdsourcing platform, Prolific, which draws mostly on Western participants, and the panels remain uneven in size. The benchmark should therefore be understood as moving closer to listener-derived evaluation across cultures.

AI Usage Statement

During original pipeline development, AI tools were used to help run the sensitivity analyses that informed the parameters reported in Methods. The authors selected and verified the final values. During final revisions, these tools also helped refine the design of additional analyses, check their implementation, and draft documentation for the released code and supplementary material. The authors verified all code and outputs and take responsibility for the paper. AI tools were also used to assist manuscript editing and proofreading. The authors reviewed each of the edit suggestions and approved the final version.

5. REFERENCES

[1] M. E. P. Davies, S. Böck, and M. Fuentes, "Tempo, beat and downbeat estimation," Tutorial at the 22nd International Society for Music Information Retrieval Conference, 2021. [Online]. Available: https://tempobeatdownbeat.github.io/tutorial/intro.html

[2] F. Foscarin, J. Schlüter, and G. Widmer, "Beat This! Accurate beat tracking without DBN postprocessing," in *Proc. ISMIR*, 2024, pp. 962–969.

[3] F. Krebs, S. Böck, and G. Widmer, "Rhythmic pattern modeling for beat and downbeat tracking in musical audio," in *Proc. ISMIR*, 2013, pp. 227–232.

[4] M. Goto, H. Hashiguchi, T. Nishimura, and R. Oka, "RWC music database: Popular, classical, and jazz music databases," in *Proc. ISMIR*, 2002, pp. 287–288.

[5] C. Harte, "Towards automatic extraction of harmony information from music signals," Ph.D. dissertation, Queen Mary University of London, Aug. 2010, source of the Isophonics Beatles annotations. [Online]. Available: https://theses.eurasip.org/document/towards-automatic-extraction-of-harmony-information-from-music-signals/

[6] O. Nieto, M. McCallum, M. E. P. Davies, A. Robertson, A. Stark, and E. Egozy, "The Harmonix Set: Beats, downbeats, and functional segment annotations of Western popular music," in *Proc. ISMIR*, 2019, pp. 565–572.

[7] A. Holzapfel, B. L. Sturm, and M. Coeckelbergh, "Ethical dimensions of music information retrieval technology," *Transactions of the International Society for Music Information Retrieval*, vol. 1, no. 1, pp. 44–55, 2018.

[8] L. O. Nunes, M. Rocamora, L. Jure, and L. W. P. Biscainho, "Beat and downbeat tracking based on rhythmic patterns applied to the Uruguayan Candombe drumming," in *Proc. ISMIR*, 2015, pp. 264–270.

[9] A. Srinivasamurthy and X. Serra, "A supervised approach to hierarchical metrical cycle tracking from audio music recordings," in *Proc. ICASSP*, 2014, pp. 5217–5221.

[10] M. Anglada-Tort, P. M. Harrison, and N. Jacoby, "REPP: A robust cross-platform solution for online sensorimotor synchronization experiments," *Behavior Research Methods*, vol. 54, no. 5, pp. 2271–2285, 2022.

[11] B. H. Repp, "Sensorimotor synchronization: A review of the tapping literature," *Psychonomic Bulletin & Review*, vol. 12, no. 6, pp. 969–992, 2005.

[12] M. F. McKinney, D. Moelants, M. E. P. Davies, and A. Klapuri, "Evaluation of audio beat tracking and music tempo extraction algorithms," *Journal of New Music Research*, vol. 36, no. 1, pp. 1–16, 2007.

[13] M. Niarchou, D. E. Gustavson, J. F. Sathirapongsasuti, M. Anglada-Tort, E. Eising, E. Bell *et al.*, "Genome-wide association study of musical beat synchronization demonstrates high polygenicity," *Nature Human Behaviour*, vol. 6, no. 9, pp. 1292–1309, 2022.

[14] N. Jacoby, R. Polak, J. A. Grahn, D. J. Cameron, K. M. Lee, R. Godoy, E. A. Undurraga, T. Huanca, T. Thalwitzer, N. Doumbia, D. Goldberg, E. H. Margulis, P. C. M. Wong, L. Jure, M. Rocamora, S. Fujii, P. E. Savage, J. Ajimi, R. Konno, S. Oishi, K. Jakubowski, A. Holzapfel, E. Mungan, E. Kaya, P. Rao, M. A. Rohit, S. Alladi, B. Tarr, M. Anglada-Tort, P. M. C. Harrison, M. J. McPherson, S. Dolan, A. Durango, and J. H. McDermott, "Commonality and variation in mental representations of music revealed by a cross-cultural comparison of rhythm priors in 15 countries," *Nature Human Behaviour*, vol. 8, no. 5, pp. 846–877, 2024.

[15] M. F. McKinney and D. Moelants, "Ambiguity in tempo perception: What draws listeners to different metrical levels?" *Music Perception*, vol. 24, no. 2, pp. 155–166, 2006.

[16] H. Lee, E. Celen, P. M. C. Harrison, M. Anglada-Tort, P. van Rijn, M. Park, M. Schönwiesner, and N. Jacoby, "GlobalMood: A cross-cultural benchmark for music emotion recognition," in *Proc. ISMIR*, 2025, pp. 11–19.

[17] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.

[18] U. Marchand, Q. Fresnel, and G. Peeters, "GTZAN-Rhythm: Extending the GTZAN test-set with beat, downbeat and swing annotations," in *Late-Breaking/Demo Session, Proc. ISMIR*, 2015, HAL preprint hal-01252607.

[19] M. Goto, "AIST annotation for the RWC music database," in *Proc. ISMIR*, 2006, pp. 359–360.

[20] J. Driedger, H. Schreiber, W. B. de Haas, and M. Müller, "Towards automatically correcting tapped beat annotations for music recordings," in *Proc. ISMIR*, 2019, pp. 200–207.

[21] S. Hainsworth and M. Macleod, "Particle filtering applied to musical tempo tracking," *EURASIP Journal on Applied Signal Processing*, vol. 2004, no. 15, pp. 2385–2395, 2004.

[22] A. Srinivasamurthy, A. Holzapfel, and X. Serra, "In search of automatic rhythm analysis methods for Turkish and Indian art music," *Journal of New Music Research*, vol. 43, no. 1, pp. 94–114, 2014.

[23] A. Holzapfel, F. Krebs, and A. Srinivasamurthy, "Tracking the "odd": Meter inference in a culturally diverse music corpus," in *Proc. ISMIR*, 2014, pp. 425–430.

[24] F. Foscarin, A. McLeod, P. Rigaux, F. Jacquemard, and M. Sakai, "ASAP: A dataset of aligned scores and performances for piano transcription," in *Proc. ISMIR*, 2020, pp. 534–541.

[25] Q. Xi, R. M. Bittner, J. Pauwels, X. Ye, and J. P. Bello, "GuitarSet: A dataset for guitar transcription," in *Proc. ISMIR*, 2018, pp. 453–460.

[26] J. Gillick, A. Roberts, J. Engel, D. Eck, and D. Bamman, "Learning to groove with inverse sequence transformations," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 2269–2279.

[27] B. H. Repp and Y.-H. Su, "Sensorimotor synchronization: A review of recent research (2006–2012)," *Psychonomic Bulletin & Review*, vol. 20, no. 3, pp. 403–452, 2013.

[28] J. Morton, S. Marcus, and C. Frankish, "Perceptual centers (P-centers)," *Psychological Review*, vol. 83, no. 5, pp. 405–408, 1976.

[29] A. Danielsen, K. Nymoen, E. Anderson, G. S. Câmara, M. T. Langerød, M. R. Thompson, and J. London, "Where is the beat in that note? Effects of attack, duration, and frequency on the perceived timing of musical and quasi-musical sounds," *Journal of Experimental Psychology: Human Perception and Performance*, vol. 45, no. 3, pp. 402–418, 2019.

[30] J. London, K. Nymoen, M. T. Langerød, M. R. Thompson, D. L. Code, and A. Danielsen, "A comparison of methods for investigating the perceptual center of musical sounds," *Attention, Perception, & Psychophysics*, vol. 81, no. 6, pp. 2088–2101, 2019.

[31] P. Toiviainen and J. S. Snyder, "Tapping to Bach: Resonance-based modeling of pulse," *Music Perception*, vol. 21, no. 1, pp. 43–80, 2003.

[32] R. B. Dannenberg and L. A. Wasserman, "Estimating the error distribution of a single tap sequence without ground truth," in *Proc. ISMIR*, 2009, pp. 297–302.

[33] M. Sadakata, P. Desain, and H. Honing, "The Bayesian way to relate rhythm perception and production," *Music Perception*, vol. 23, no. 3, pp. 269–288, 2006.

[34] N. Jacoby and J. H. McDermott, "Integer ratio priors on musical rhythm revealed cross-culturally by iterated reproduction," *Current Biology*, vol. 27, no. 3, pp. 359–370, 2017.

[35] D. Müllensiefen, B. Gingras, J. Musil, and L. Stewart, "The musicality of non-musicians: An index for assessing musical sophistication in the general population," *PLOS ONE*, vol. 9, no. 2, p. e89642, 2014.

[36] P. Harrison, R. Marjieh, F. Adolfi, P. van Rijn, M. Anglada-Tort, O. Tchernichovski, P. Larrouy-Maestri, and N. Jacoby, "Gibbs sampling with people," *Advances in Neural Information Processing Systems*, vol. 33, pp. 10 659–10 671, 2020.

[37] J. R. Iversen and A. D. Patel, "The beat alignment test (BAT): Surveying beat processing abilities in the general population," in *Proceedings of the 10th International Conference on Music Perception and Cognition*, Sapporo, Japan, 2008, pp. 465–468.

[38] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.

[39] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, and D. P. W. Ellis, "mir_eval: A transparent implementation of common MIR metrics," in *Proc. ISMIR*, 2014, pp. 367–372.

[40] N. Singer, N. Jacoby, T. Hendler, and R. Granot, "Feeling the beat: Temporal predictability is associated with ongoing changes in music-induced pleasantness," *Journal of Cognition*, vol. 6, no. 1, p. 34, 2023.

[41] S. Böck, F. Krebs, and G. Widmer, "A multi-model approach to beat tracking considering heterogeneous music styles," in *Proc. ISMIR*, 2014, pp. 603–608.

[42] M. E. P. Davies, N. Degara, and M. D. Plumbley, "Evaluation methods for musical audio beat tracking algorithms," Queen Mary University of London, Centre for Digital Music, Tech. Rep. C4DM-TR-09-06, 2009.

[43] D. J. Cameron, J. Bentley, and J. A. Grahn, "Cross-cultural influences on rhythm processing: reproduction, discrimination, and beat tapping," *Frontiers in Psychology*, vol. 6, p. 366, 2015.

[44] E. E. Hannon, G. Soley, and S. Ullal, "Familiarity overrides complexity in rhythm perception: a cross-cultural comparison of American and Turkish listeners," *Journal of Experimental Psychology: Human Perception and Performance*, vol. 38, no. 3, pp. 543–548, 2012.

[45] K. Jakubowski, R. Polak, M. Rocamora, L. Jure, and N. Jacoby, "Aesthetics of musical timing: Culture and expertise affect preferences for isochrony but not synchrony," *Cognition*, vol. 227, p. 105205, 2022.

[46] C. Drake, A. Penel, and E. Bigand, "Tapping in time with mechanically and expressively performed music," *Music Perception*, vol. 18, no. 1, pp. 1–23, 2000.

[47] P. van Rijn, Y. Sun, H. Lee, R. Marjieh, I. Sucholutsky, F. Lanzarini, E. André, and N. Jacoby, "One test, many tongues: Surveying language proficiency across the globe," *Proceedings of the National Academy of Sciences*, vol. 123, no. 13, p. e2420179123, 2026.

[48] B. H. Repp, "Control of expressive and metronomic timing in pianists," *Journal of Motor Behavior*, vol. 31, no. 2, pp. 145–164, 1999.

[49] P. E. Savage, N. Jacoby, E. H. Margulis, H. Daikoku, M. Anglada-Tort, S. E.-S. Castelo-Branco, F. E. Nweke, S. Fujii, S. Hegde, Hu Chuan-Peng, J. Jabbour, C. Lew-Williams, D. Mangalagiu, R. McNamara, D. Müllensiefen, P. Opondo, A. D. Patel, and H. Schippers, "Building sustainable global collaborative networks: Recommendations from music studies and the social sciences," in *The Science-Music Borderlands*. The MIT Press, 2023, pp. 347–366.

[50] N. Jacoby, E. H. Margulis, M. Clayton, E. Hannon, H. Honing, J. Iversen, T. R. Klein, S. A. Mehr, L. Pearson, I. Peretz *et al.*, "Cross-cultural work in music cognition: Challenges, insights, and recommendations," *Music Perception*, vol. 37, no. 3, pp. 185–195, 2020.

A. SUPPLEMENTARY ANALYSES

This appendix contains the supplementary figures, tables, and analyses. Playhead videos of the patterns in Section 3.6 are in the supplementary HTML, and the underlying audio and video files are in the companion repository.

A.1 Per-corpus rating distributions

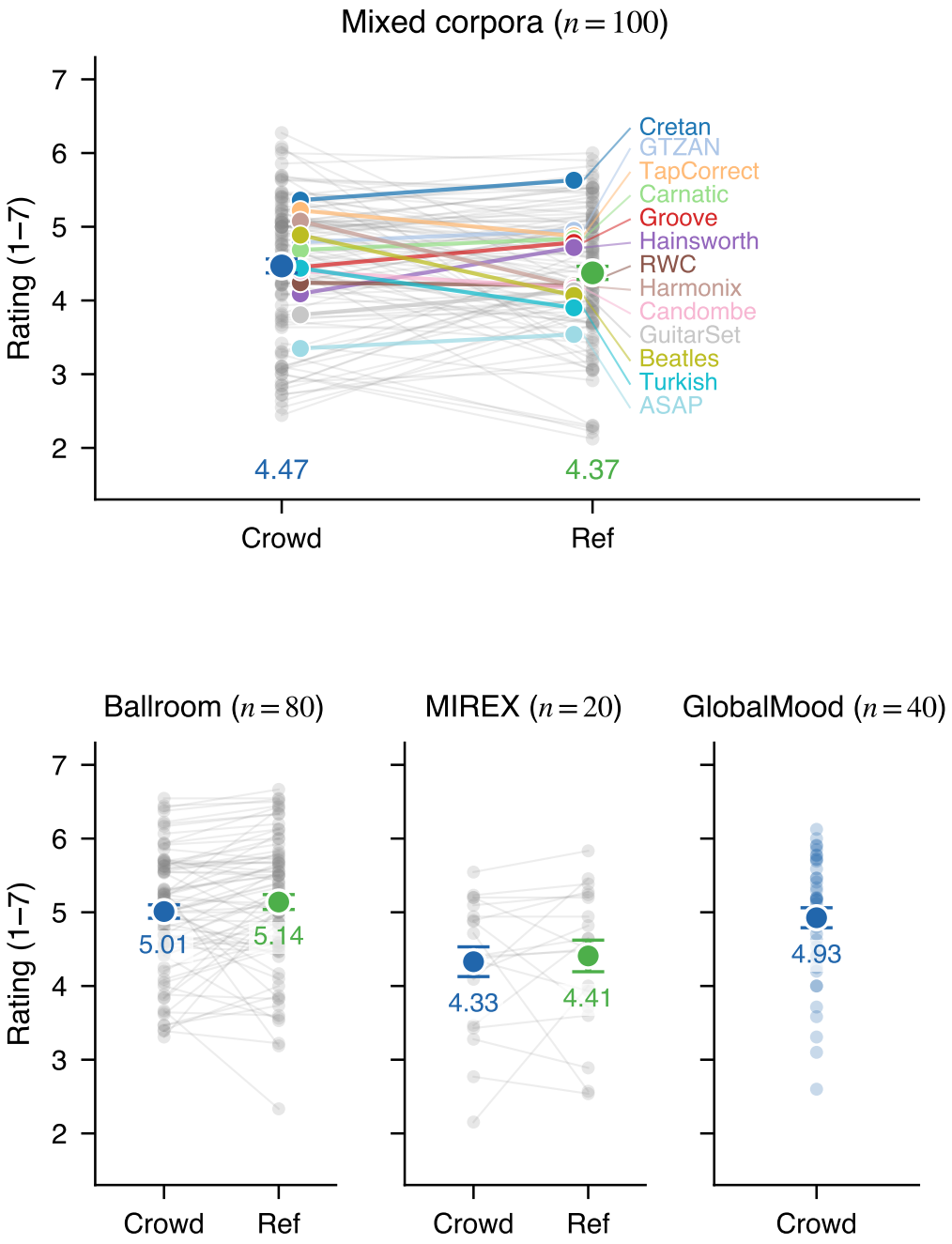

Figure A1. Per-corpus distributions of mean per-excerpt ratings on the 1–7 scale for Crowd and Reference annotations across the four validation corpus groups. The summary statistics from this figure (per-corpus means, Δ, p-values, and effect sizes) are reported in Table 1. The mixed-corpora plot also shows the per-source Crowd and Reference means for its 13 constituent corpora. Error bars are ± 1 SE across excerpts.

A.2 Per-source results for the mixed corpora

The 100 mixed-corpora excerpts come from 13 source corpora. Wilcoxon signed-rank tests compare excerpt-level Crowd and Reference means within each source, with Benjamini–Hochberg correction across the 13 tests. None remain significant at $\alpha = .05$ after correction.

Table A1. Crowd and Reference ratings by source corpus in the mixed corpora. Crowd and Ref are mean per-excerpt ratings. Δ = Crowd − Ref. p_{BH} is corrected across the 13 source-corpus tests.

Source corpus	n	Crowd	Ref	Δ	p	p_{BH}	d_z
ASAP	8	3.35	3.54	-0.19	.742	.877	-0.20
Beatles	8	4.89	4.07	$+0.82$	.055	.178	$+0.83$
Candombe	13	4.40	4.17	$+0.23$	.733	.877	$+0.20$
Carnatic	4	4.69	4.84	-0.15	.625	.877	-0.30
Cretan	3	5.36	5.63	-0.27	.500	.812	-0.80
Groove (MIDI)	8	4.45	4.79	-0.33	.312	.580	-0.44
GTZAN	8	4.79	4.95	-0.16	.844	.914	-0.23
GuitarSet	8	3.81	4.13	-0.32	.031	.135	-0.75
Hainsworth	8	4.09	4.72	-0.63	.008	.102	-1.31
Harmonix	8	5.08	4.20	$+0.88$	.078	.203	$+0.74$
RWC	8	4.24	4.20	$+0.04$	.938	.938	$+0.04$
TapCorrect	8	5.22	4.88	$+0.34$	.250	.542	$+0.44$
Turkish	8	4.44	3.90	$+0.54$	.016	.102	$+1.39$

A.3 Participant demographics and study flow

Benchmark recruitment and exclusions. Across the ten country panels, 2,160 people began a session: 707 completed it, 831 did not pass the tapping performance check, 134 declined a consent step, and 488 left before finishing. Of the 28,945 recorded trials, 23,493 passed the recording, marker-analysis, and post-trial checks. The 746 benchmark tappers are those with at least one such trial (698 who completed and 48 who left early). Of the 139 people who started the listening experiment, 130 finished it, and all 6,487 reported ratings come from them.

Table A2. Tapper groups in the benchmark and validation analyses.

Excerpt set	Reference source	Excerpts	Tappers	Median per excerpt	Trials
Ballroom	Expert beat grids	80	231	40	3,821
Mixed corpora	13 source corpora	100	104	36	3,555
MIREX	Lab-tapper consensus (40 participants)	20	102	67	1,353
GlobalMood validation	None (absolute-quality check)	40	296	110	4,403
GlobalTap benchmark	Crowd-derived	200	746	117	23,493
Total	—	440	1,479	—	36,625

The trial totals and median number of tappers per excerpt include only trials that passed the recording, marker, and post-trial checks.

Table A3. Demographics of the 746 GlobalTap benchmark tappers.

Variable	Distribution
Age (years)	$M = 33.1$, $SD = 11.1$
Gender	49.5% female, 49.1% male, 1.4% non-binary or undisclosed
Formal musical training	Median 1 year (IQR 0–4), with 33% none and 9% at least 10 years
Training median by panel	1–6 years
Gold-MSI (25-item, 1–7)	Overall $M = 4.10$, $SD = 0.88$, with panel means 3.77–4.56

A.4 Tapper-count convergence

Figure A2 shows convergence of participant-normalized tapping KDEs and their extracted peaks. Random subsamples of tappers were compared with the full consensus they were drawn from. Density correlation reached .90 at 40 tappers and peak F-measure reached .90 at 30 tappers. At 90 tappers, the values were .985 and .971.

The independent-group comparison in §2.3 ran two separate groups per excerpt through KDE, peak extraction, and optimization. Beyond the final-grid F-measures reported there, the two groups' grids had the same beat rate on 85.0% of excerpts at $N = 20$ and 88.5% at $N = 50$.

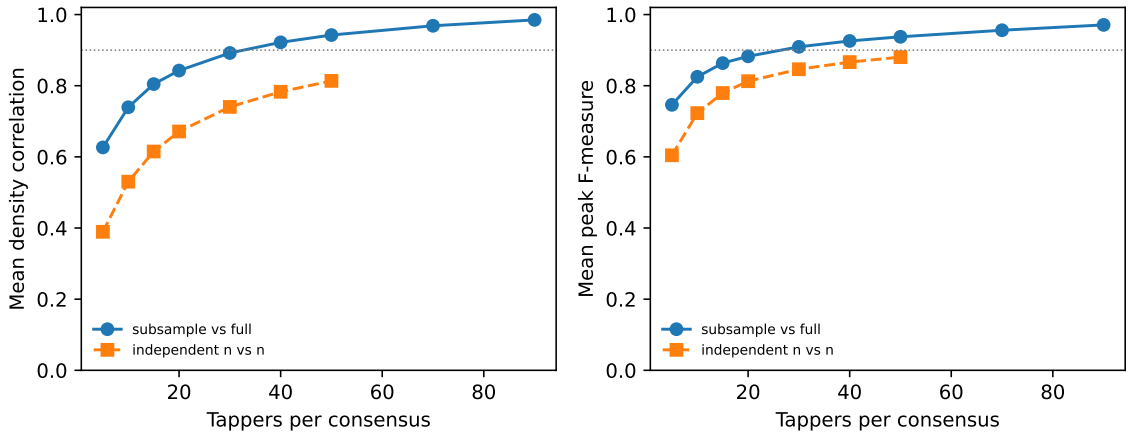

Figure A2. Convergence as a function of tappers per excerpt. Circles compare subsamples with the full consensus. Squares compare independent groups. The dotted horizontal line marks .90.

A.5 Exploratory country-panel analyses

For each excerpt, each trial's tapping period was compared with a reference period, the median of the other nine panels' median periods, and classified as double rate (ratio .35–.65), matched rate (.80–1.25), half rate (1.75–2.25), or other/unclear. The period analysis used 23,492 trials (one of the 23,493 had too few taps for a period estimate) and the phase analysis the 11,450 matched-rate trials. Panels were compared by subtracting each excerpt's mean from every trial's outcome, averaging the remainder within each participant, and permuting panel labels across participants (9,999 permutations).

For the own-country test in the five focal countries, ISTC grows with the number of tappers, so for each excerpt and panel it was averaged over 500 random samples of seven tappers, the most available for every excerpt and panel. The regression had a separate baseline for each excerpt and each panel, with standard errors clustered by excerpt.

A.6 Metrical-preference analysis

This analysis includes all 180 excerpts with independent expert beat references. Excerpts are grouped by the Crowd annotation's period relative to the reference. Rating differences are excerpt-level Crowd minus Reference means. An excerpt is classified as matched, double, or half rate when its crowd-to-reference period ratio falls within ± 0.15 octaves (about $\pm 11\%$) of $1\times$, $0.5\times$, or $2\times$, respectively. All remaining excerpts are grouped as other ratios.

On matched-rate excerpts, an excerpt's phase offset is the median signed distance from each Crowd beat to the nearest reference beat within 150 ms, over the rendered 10–25 s window, and it counts as consistent at 20 ms or more. On 8 excerpts no Crowd beat came within 150 ms of a reference beat: the Crowd beats fell a third to half a beat from the reference's. These also count as consistent offsets.

Table A4. Rating differences by Crowd-to-Reference period relation.

Period relation	n	Crowd higher	Median Δ
Matched rate	145	51	$-.143$
Crowd half rate	9	9	$+.748$
Crowd double rate	14	7	$-.238$
Other ratio	12	7	$+.393$

A.7 Beat metrics beyond F-measure

Table A5 reports standard `mir_eval` metrics for the 198 excerpts evaluable for both trackers.

Table A5. Beat-tracking agreement with the GlobalTap consensus.

Metric	madmom	Beat This!
CMLc	.797	.621
CMLt	.807	.636
AMLc	.947	.908
AMLt	.956	.935
Information gain	.669	.643

Paired excerpt-level tests favored madmom for all five metrics, with FDR correction across all ten `mir_eval` metrics (the other five are in the repository). CMLc, CMLt, and information gain reached $q < .001$, AMLc $q = .011$, and AMLt $q = .024$. Pooled (pre-optimization) ISTC was associated with agreement for madmom ($\rho = .39$, $p < .0001$) and Beat

This! ($\rho = .27$, $p = .0001$). Pipeline-output CV was also associated with agreement for Beat This! ($\rho = -.24$, $p = .0006$) and madmom ($\rho = -.18$, $p = .009$). The madmom correlations use all 200 excerpts and the Beat This! correlations the 198 it could evaluate.

A.8 Grid optimizer settings

Table A6 lists the settings of the grid optimizer in §2.3. T is the grid period. The companion repository describes every setting with the code (`supplementary_analyses/METHOD_PARAMETERS.md`).

Table A6. Grid optimizer settings.

Step	Setting
Search	Period 0.28–1.25 s (48–214 BPM) in 120 steps, phase in 240 steps per period. C is the proportion of peaks within $0.12T + 20$ ms of a grid beat. The density has an 80 ms Gaussian at each peak, scaled to a maximum of 1
Refinement	Least squares on peaks within $0.15T$ of the grid, then the phase re-searched within $\pm 0.1T$, keeping only phases in $[0, T)$
Snapping	A beat with a peak within $0.12T$ moves 45% of the way toward it
Rate doubling	When more than 55% of the midpoints between beats have a peak within $0.18T$, the mean density at the midpoints is over 40% of that at the beats, and the doubled rate is at most 214 BPM. The beats of the doubled grid are then nudged toward even spacing
Gap filling	Beats inserted in gaps over $1.4T$ and out to the window edges, at the median beat interval
Split peaks	Tried when the largest gap between the sorted inter-peak intervals shorter than $0.9T$ exceeds $0.10T$. The grids are compared by $2C' + (1 - \mathrm{CV}) + 0.5\,\bar{D}$, where C' is the proportion of peaks within $0.12T + 30$ ms of a beat and CV is the coefficient of variation of the beat intervals